Tugas Kecil 3 IF3058 Kriptografi

Pembuatan Applet Java yang Mengimplementasikan fungsi SHA1

Anggota kelompok:

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1. Source Code Program

Kelas SHA1.Java

```
package sha;

import java.io.BufferedReader;
import java.io.DataInputStream;
import java.io.File;
import java.io.FileInputStream;
import java.io.FileNotFoundException;
import java.io.IOException;
import java.io.InputStreamReader;
import java.util.logging.Level;
import java.util.logging.Logger;

public class SHA1 {
    private final int[] K = {0x5a827999, 0x6ed9eba1, 0x8f1bbcdc,
0xca62c1d6};
    public String MessageDigest;

    public SHA1(File input) {
        FileInputStream fstream = null;
        try {
            fstream = new FileInputStream(input.getAbsolutePath());
            DataInputStream in = new DataInputStream(fstream);
            BufferedReader br = new BufferedReader(new
InputStreamReader(in));
            String strLine;
            String temp = "";
            while((strLine = br.readLine()) != null) {
                temp += strLine;
                System.out.println(strLine);
            }
            SHA1 a = new SHA1(temp);
            this.MessageDigest = a.getMessageDigest();
        } catch (IOException ex) {
            Logger.getLogger(SHA1.class.getName()).log(Level.SEVERE, null,
ex);
        }
        finally {
            try {
                fstream.close();
            } catch (IOException ex) {
                Logger.getLogger(SHA1.class.getName()).log(Level.SEVERE,
null, ex);
            }
        }
    }
```

```
    public String getMessageDigest(){
        return MessageDigest;
    }

    public static void main(String []args)
   {
        String sTest= "abc";
        SHA1 a;
        a = new SHA1(sTest);
        System.out.println(a.getMessageDigest());
   }

    public SHA1(String input){
        input += (char) 0x80;

        int p = input.length()/4 + 2;
        int N = (int) Math.ceil(p/16);
        N++;
        int[][] M = new int[N][];
        for (int i=0; i<N; i++) {
            M[i] = new int[16];
            for (int j=0; j<16; j++) {
                int a,b,c,d;
                if(i*64+j*4 >= input.length()) a = 0;
                else a = (input.charAt(i*64+j*4)<<24);
                if(i*64+j*4+1 >= input.length()) b = 0;
                else b = (input.charAt(i*64+j*4+1)<<16);
                if(i*64+j*4+2 >= input.length()) c = 0;
                else c = (input.charAt(i*64+j*4+2)<<8);
                if(i*64+j*4+3 >= input.length()) d = 0;
                else d = (input.charAt(i*64+j*4+3));
                M[i][j] = a | b | c | d;
            }
        }

        M[N-1][14] = (int) (((input.length() - 1) * 8) / Math.pow(2, 32));
        M[N-1][14] = (int) Math.floor(M[N-1][14]);
        M[N-1][15] = ((input.length()-1)*8) & 0xffffffff;

        int H0 = 0x67452301;
        int H1 = 0xefcdab89;
        int H2 = 0x98badcfe;
        int H3 = 0x10325476;
        int H4 = 0xc3d2e1f0;

        int[] W = new int[80];
        int a, b, c, d, e;
        for (int i=0; i<N; i++) {

            for (int t=0;  t<16; t++) W[t] = M[i][t];
            for (int t=16; t<80; t++) W[t] = rotation(W[t-3] ^ W[t-8] ^
W[t-14] ^ W[t-16], 1);


            a = H0; b = H1; c = H2; d = H3; e = H4;


            for (int t=0; t<80; t++) {
                int s = (int) Math.floor(t/20);
                int T = (rotation(a,5) + function(s,b,c,d) + e + K[s] +
W[t]) & 0xffffffff;
                e = d;
                d = c;
```

```
                c = rotation(b, 30);
                b = a;
                a = T;
            }

            H0 = (H0+a) & 0xffffffff;
            H1 = (H1+b) & 0xffffffff;
            H2 = (H2+c) & 0xffffffff;
            H3 = (H3+d) & 0xffffffff;
            H4 = (H4+e) & 0xffffffff;
        }

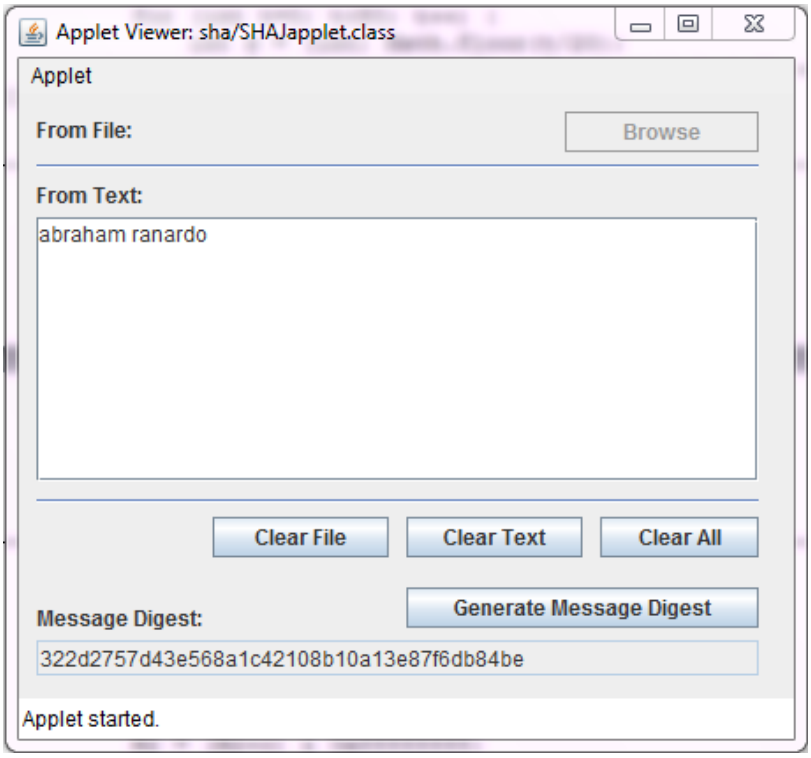
        MessageDigest = inttoHexStr(H0) + inttoHexStr(H1) + inttoHexStr(H2)
+ inttoHexStr(H3) + inttoHexStr(H4);
    }

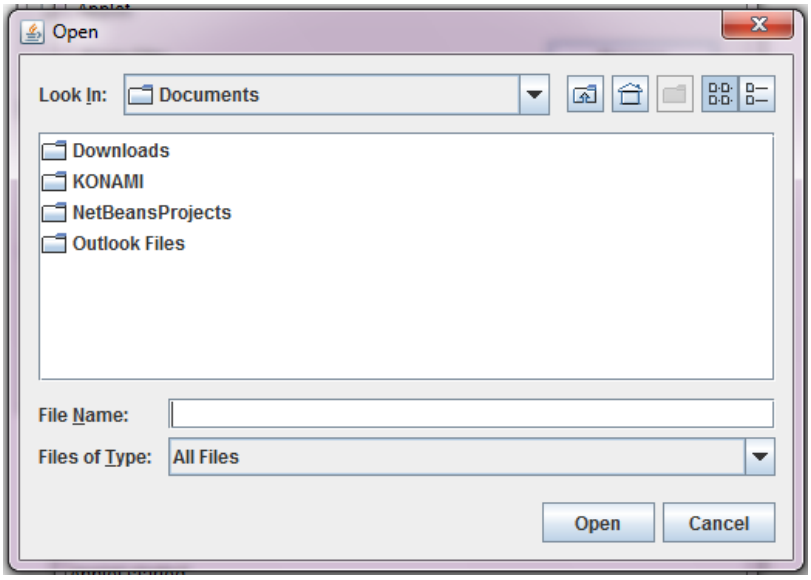
    private int function(int s, int x, int y, int z) {
        switch (s) {
            case 0: return (x & y) ^ (~x & z);
            case 1: return x ^ y ^ z;
            case 2: return (x & y) ^ (x & z) ^ (y & z);
            case 3: return x ^ y ^ z;
            default: return 0;
        }
    }

    private String inttoHexStr(int a){
        String s="";
        int v;
        for (int i=7; i>=0; i--) {
            v = (a>>>(i*4)) & 0xf;
            s += Integer.toHexString(v);
        }
        return s;
    }

    private int rotation(int x, int n){
        return (x<<n) | (x>>>(32-n));
    }
}
```

2. Antarmuka Program





3. Contoh Pengujian

Pesan (small): abraham ranardo Message Digest: 322d2757d43e568a1c42108b10a13e87f6db84be

Pesan (medium): Use this item to permanently increase your health bar. Using it will actually completely refill your health, so I recommend holding on to these and using them during particularly challenging boss fights. Message Digest: 7e063955742a5ba80dd691f29125aa955e407a73

Pesan (large): You will see a hole in a wall directly in front of where you started. Press the Right Bumper to open it for another scene. Afterwards, the camera controls will be displayed below. Return to the shop and open the chest in front of the counter which contains an Herb of Spiritual Life (Acolyte Only). You can break the barrels outside of the shop for additional Yellow Essence. Continue down the path until you get a short tutorial on the Earth Dragon Statues. You can save and heal at these when no enemies are in the area. After saving, four enemies will come from the other side of the walkway and attack. Take care of them however you see fit. Continue past the shop until you come to the now opened red gate. Enter through the gate and walk down the stairs and across the platform until you see a door with a chest on the left side. This chest contains the Reverse Wind Technique. This is basically a quick dash. Hold the Left Trigger then push the left stick in the direction you wish to dash to perform it. Now kick open the door and take out the four Spider Clan Ninjas that attempt to ambush you. Continue down the hallway. The chest at the end contains Blue Essence which restores your health. Go down the other hallway in this V shaped area. Four more ninjas will run across the wall, just to be stopped by a gory death. Take the ladder down into the small room below. There is a chest here which contains The Technique of Shadowless Footsteps. Press A to jump at a wall and hold the direction you wish to wall run with the Left Stick. Message Digest: a4541b43e51826540467e9392a57e3f9740c7cb0

Apakah program sensitif terhadap perubahan satu karakter?

Misalnya pesan awal ialah abraham ranardo diubah menjadi pesan baru, yaitu abraham Ranardo.

Maka Message Digestnya ialah e8661d1755363debcfcfde2cc3ad42352079695c. **Sensitif** terhadap perubahan satu karakter!